

Grace — over-determination pass over the PD tier (2026-08-03, K1137)

Question: which PD entries are UNDER-claimed because a forced structure predicts MANY data points on FEW parameters (over-determined = real evidence beyond a one-number PD)? Calibrate BOTH ways — willing to return “none.”

Discipline: over-determination raises the CONFIRMATION axis (structure corroborated), it does NOT force an asserted strength → it can strengthen a PD’s standing but NOT promote PD→Derived. Genuine over-determination = N independent data points / $k < N$ free params, STRUCTURE (not integers) the constraint (archetype: binding ladder T2540, 6 pts / 2 params).

The 9 PD entries, assessed

entry	data pts	free params	over-determined?	verdict
n_s (const_016)	1 (0.965)	mode-magnitude 1/N_max	NO — one number	correctly PD (ratified #7 today); NOT under-claimed correctly PD;
Ω_Λ (const_014)	1 effective (13/19)	combination	NO — “three routes” = rich-vocab of denom 19	mechanism prose OVER-claims (flagged) correctly PD
Ω_m (const_018)	0 indep (=1− Ω_Λ)	—	NO — complement	
$\kappa_{ls} \times 2 +$ magics $\times 4$ (25/81/94- 97)	7 shells / 1 κ	κ strength	YES — but Mayer–Jensen textbook	confirmation HIGH, tier PD (BST value asserted)

Findings (both directions)

1. NO tier promotions warranted. The one genuine binding-ladder-style over-determination in the corpus (T2540) is ALREADY at Structure-Derived. Looked for under-claims (flattering direction); discipline supports none.
2. Magic-number cluster IS over-determined — one spin-orbit strength opens {28,50,82,126} + kills {40,70,112} = 7 shell constraints / 1 param. BUT that over-determination is Mayer–Jensen (1949) textbook shell-closure, which BST *inherits* by identifying block+packing — it corroborates the STRUCTURE (raise confirmation), it does NOT force the BST value $\kappa=1/(2C_2)$. So: confirmation-high, tier-PD. Not a promotion.
3. Ω_Λ over-claim found (opposite direction). “13/19 via three independent routes / most overdetermined result in BST” = rich-vocabulary of denominator 19 (F417 / K741-retraction class). Flagged in tier_review; “most overdetermined” must not travel externally. Numerator $c_3=13$ forced; tier PD stays.

Recommendation to Keeper/Casey

- No PD→Derived moves. The pass’s honest yield is one OVER-claim caught (Ω_Λ language), not an under-claim — the calibrate-both-ways result.
- Optional: add a “confirmation” annotation to the magic-number cluster noting the shell-closure over-determination is textbook-inherited (high confirmation, PD tier).
- The over-determination principle (K1135) is real but rare — it applies cleanly to the binding ladder and essentially nowhere else in the current PD tier.